Status	Finished
Started	Thursday, 30 October 2025, 1:31 PM
Completed	Thursday, 30 October 2025, 1:45 PM
Duration	14 mins 13 secs
Marks	2.00/3.00
Grade	**2.00** out of 3.00 (**66.68**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Incorrect

Mark 0.00 out of 1.00

Let A and X be the sets

$$A = \{a, b, c, d, e, f\}$$

$$X = \{!, +, \text{-}\}\,.$$

In this problem we consider words written in the alphabet which is the union of the two sets: $A \cup X$. This means that a word in this alphabet can contain letters from both A and X.

However, not all words are valid. The following rules apply:

1. A word must start with a letter from the set A.
2. A word must end with a letter from the set A.
3. A word can not contain two adjacent letters from the set X.

For example the word **a+baa!c** is valid, but the words **a+baa+!c**, **!aaa+c** and **acca!** are not.

The length of a word is the number of letters it consists of. For example, the length of **a+baa!c** is 7.

Let a_n equal the number of valid words of length n.

a) Find a recurrence relation satisfied by a_n:

a!

Your last answer was interpreted as follows:

$$a!$$

The variables found in your answer were: $[a]$

Syntax help: To express the symbol "a_n" you should enter **a[n]**. For example, if you think the correct answer is $a_{n+1} - 2a_{n-3} = 1$, you should answer **a[n+1] - 2*a[n-3] = 1**.

b) Compute the value

a_1 =

c) Compute the value

a_3 =

d) Find the general solution to the reccurence relation satisfied by a_n.

a_n =

Syntax help: You are free to use constants in your answer. For example, if you think the correct answer is $2 \cdot n - A \cdot 3^n + B$, you should answer **2*n - A*3^n + B**.

Question **2**

Correct

Mark 1.00 out of 1.00

In this problem you are tasked with computing with elements in the symmetric group of permutations, $(S_5, \cdot)$.

Syntax guide: The answer to each task is an element of the group S_5 and should be given as a 2×5 matrix. E.g. if you think the answer to a question is the permutation denoted by

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

then you should answer

```
1 2 3 4 5
2 3 1 5 4
```

a)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 3 & 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 4 & 3 & 5 \end{bmatrix} =$$

12345
25413

Your last answer was interpreted as follows:

$$\begin{bmatrix} 12345 \\ 25413 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 3 & 2 & 5 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 1 & 3 & 5 & 2 \end{bmatrix} =$$

12345
53124

Your last answer was interpreted as follows:

$$\begin{bmatrix} 12345 \\ 53124 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 3 & 2 & 4 \end{bmatrix}^{-1} =$$

12345
14352

Your last answer was interpreted as follows:

$$\begin{bmatrix} 12345 \\ 14352 \end{bmatrix}$$

d)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 4 & 3 & 5 \end{bmatrix}^{-1} =$$

12345
21435

Your last answer was interpreted as follows:

$$\begin{bmatrix} 12345 \\ 21435 \end{bmatrix}$$

Question **3**

Correct

Mark 1.00 out of 1.00

Let $(P_9, \bullet_9)$ be the group of natural numbers coprime to 9 with group operation given by multiplication modulo 9. The multiplication table for this group looks like

$$\begin{bmatrix} 1 & 2 & 4 & 5 & 7 & 8 \\ 2 & . & . & . & 5 & 7 \\ 4 & 8 & 7 & 2 & 1 & 5 \\ 5 & 1 & 2 & 7 & 8 & 4 \\ 7 & . & . & . & 4 & 2 \\ 8 & . & . & . & 2 & 1 \end{bmatrix}$$

However, we have left out 9 entries in the places marked by a dot "."

Compute the missing entries in the group multiplication table:

$7 \bullet_9 5$ = 8

$2 \bullet_9 5$ = 1

$8 \bullet_9 5$ = 4

$7 \bullet_9 2$ = 5

$2 \bullet_9 2$ = 4

$8 \bullet_9 2$ = 7

$7 \bullet_9 4$ = 1

$2 \bullet_9 4$ = 8

$8 \bullet_9 4$ = 5

Your last answer was interpreted as follows:

$$8$$

Your last answer was interpreted as follows:

$$1$$

Your last answer was interpreted as follows:

$$4$$

Your last answer was interpreted as follows:

$$5$$

Your last answer was interpreted as follows:

$$4$$

Your last answer was interpreted as follows:

$$7$$

Your last answer was interpreted as follows:

$$1$$

Your last answer was interpreted as follows:

$$8$$

Your last answer was interpreted as follows:

$$5$$

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 187

Signing code: 12133

Your last answer was interpreted as follows:

$$12133$$

◄ Test 2 (topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations)

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